STRATEGIC PROPERTY INTELLIGENCE

Analysis of the Dutch housing market: Pricing, the impact of the WOZ value and structural driver

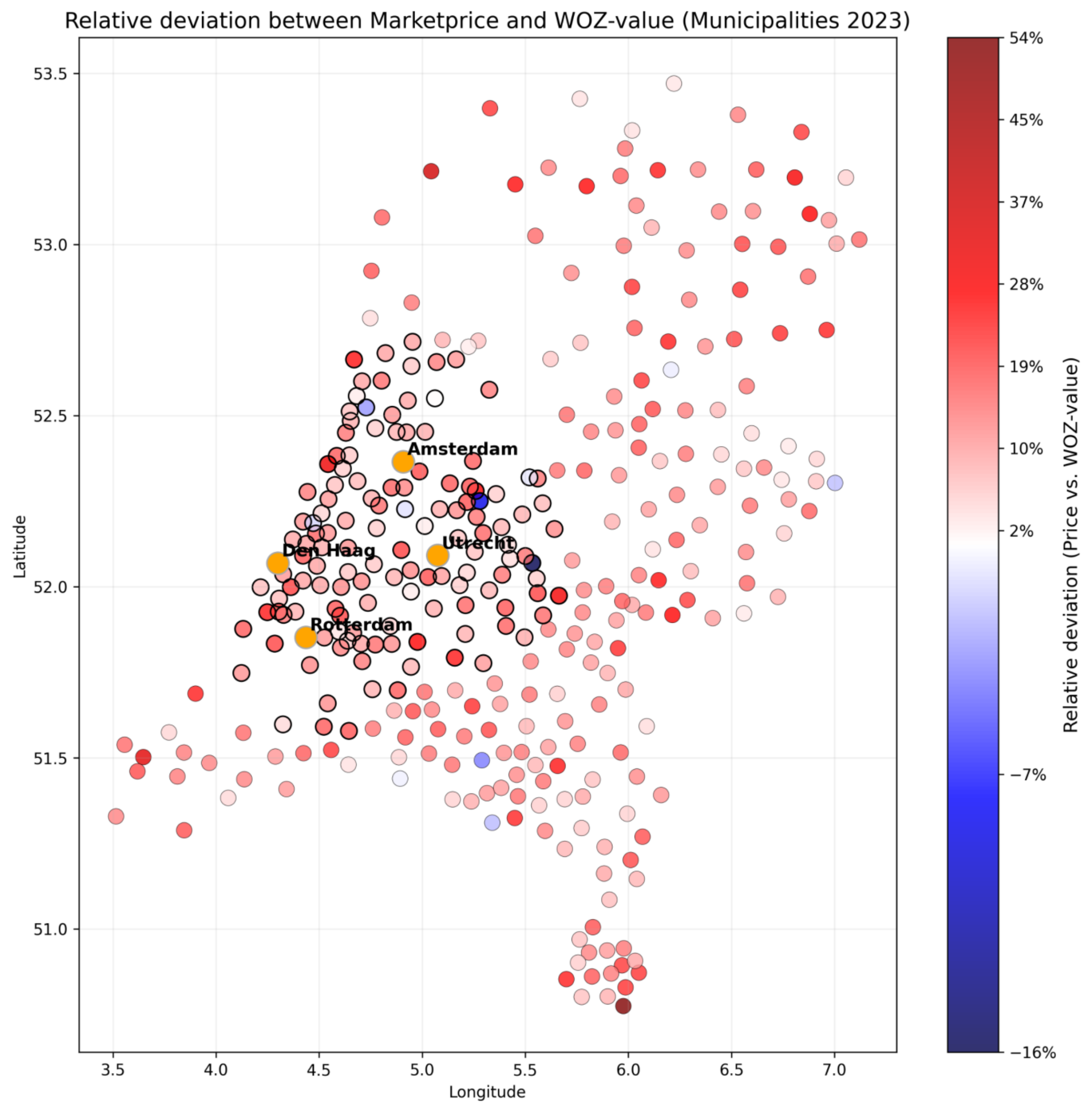
De drijfveren achter de prijsontwikkelingen
op de Nederlandse woningmarkt

CREATED WITH
Python

Overview

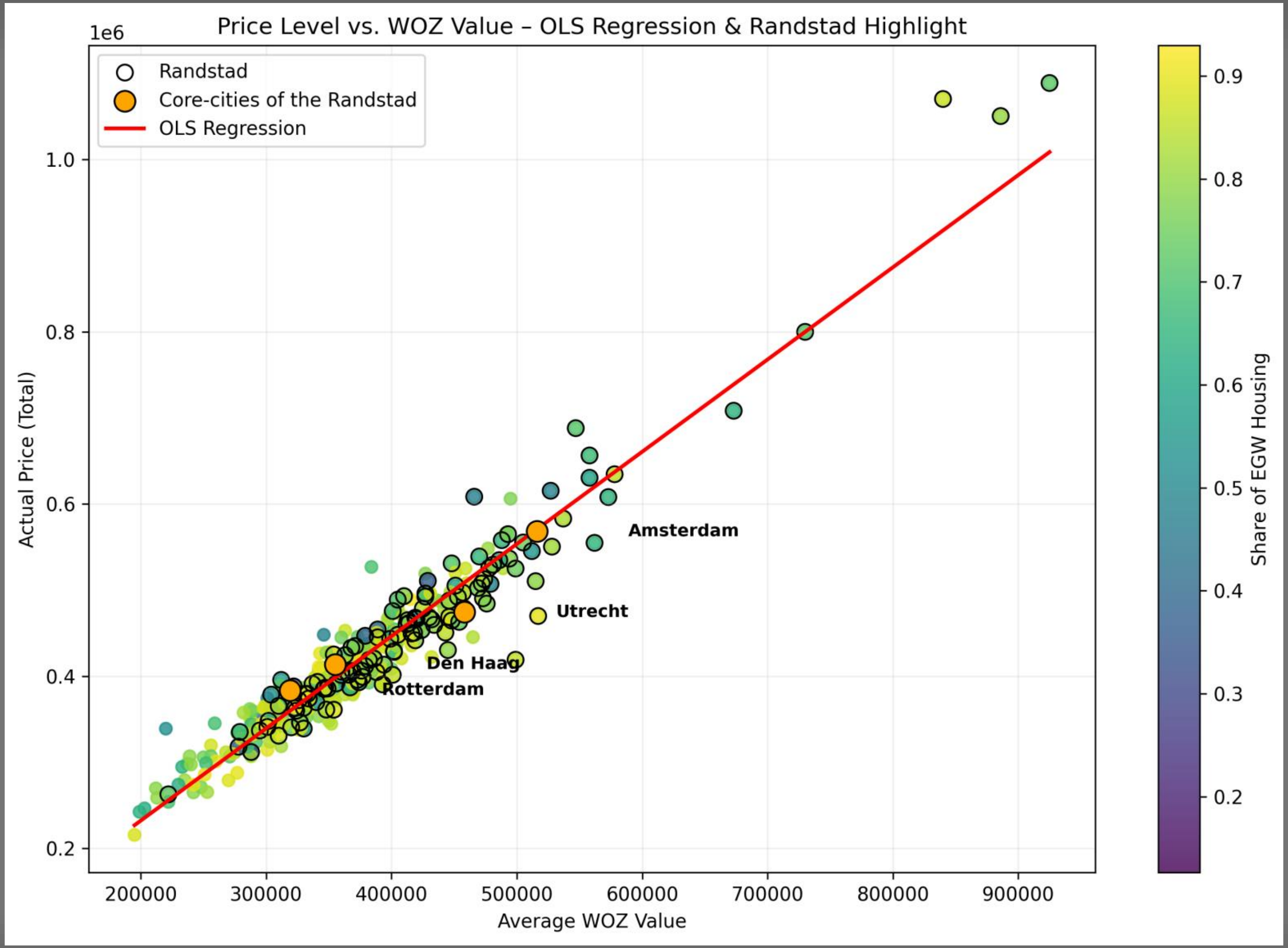
- Objective: To understand price formation in the Dutch housing market
- Data source: Centraal Bureau voor de Statistiek (**CBS**)
- Focus: Dominance of the ‘Waardering onroerende zaken’ (**WOZ**)

- Single-family houses (**EGW**) and multi-family houses (**MGW**) as distinct market segments
- Role of the Randstad as a dynamic sub-market in the Netherlands
- Limitations of the analysis due to aggregation at municipal level



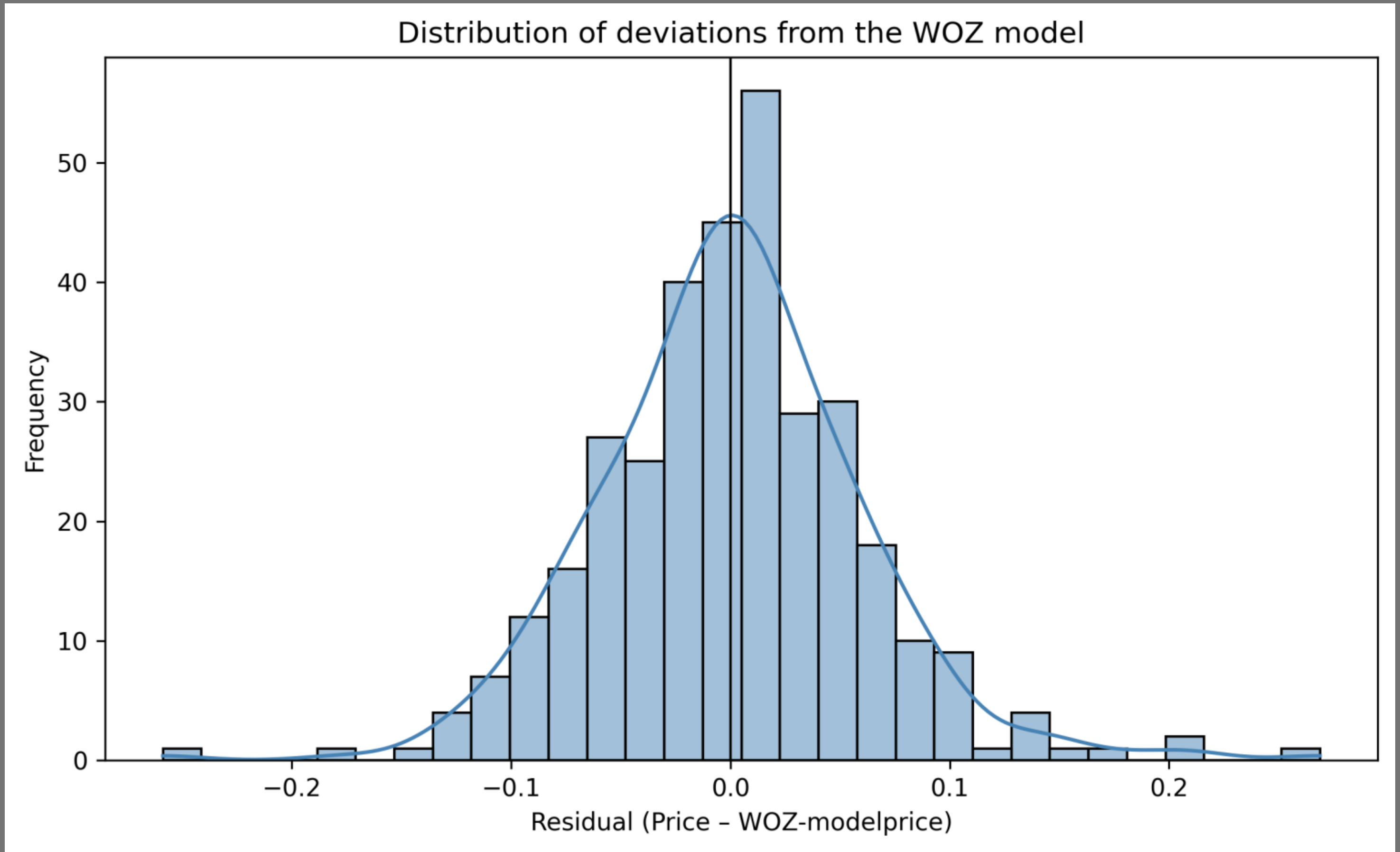
1. The WOZ-value is the dominant price driver and explains almost everything

- WOZ is the strongest predictor in all OLS models
- R^2 ranges from **0.93 to 0.97**
- Structural variables provide only minor adjustments
- WOZ is transaction-based → already provides a good representation of market value
- A one-year time lag explains slight deviations



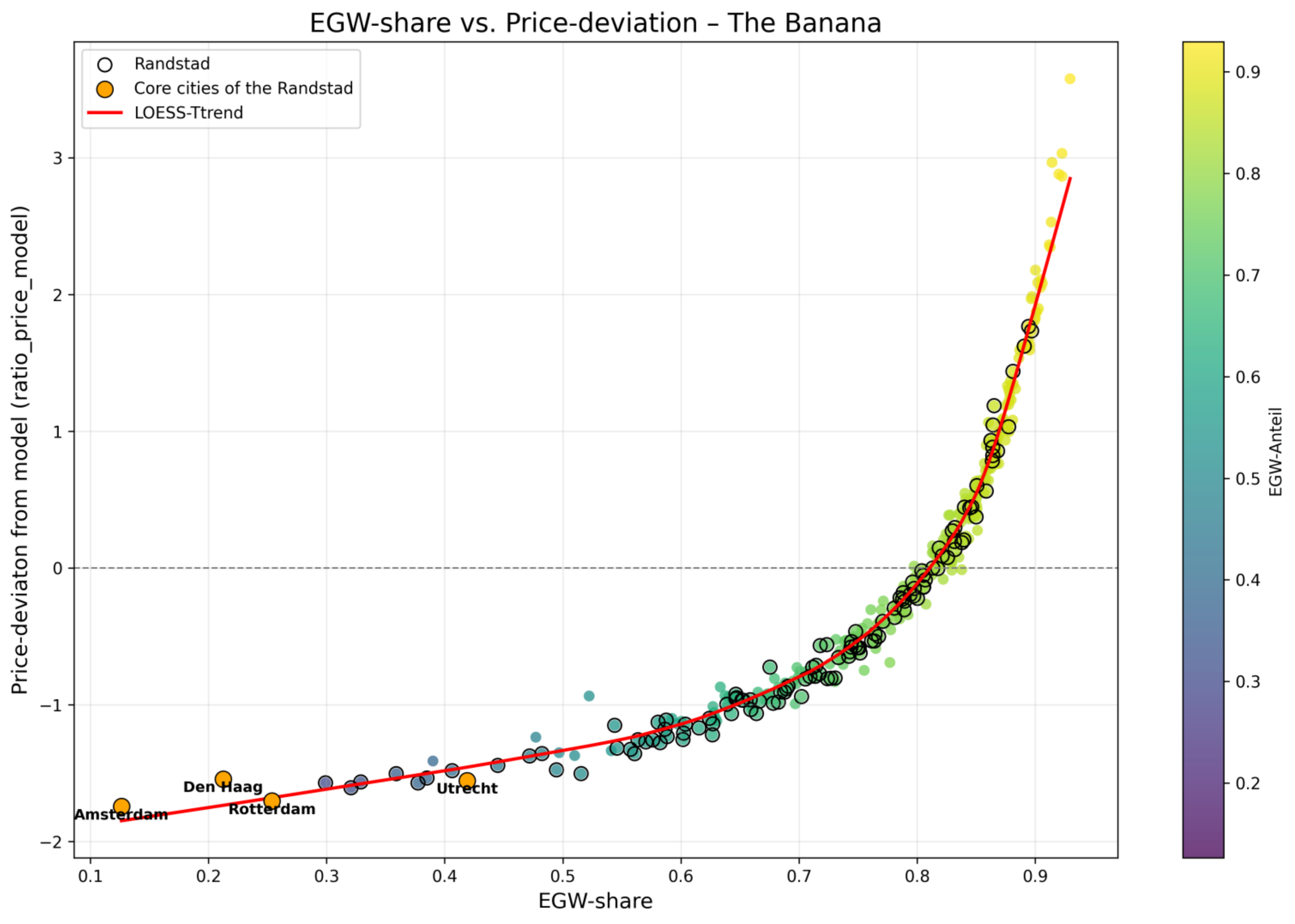
2. Structure and location variables explain deviations – but only to a limited extent

- Structural and group characteristics (e.g. proportion of EGW/MGW) are economically plausible
- Income, population density and employment are statistically significant but difficult to interpret
- Residuals show regional patterns
- Scatterplots of the structural variables: weak or diffuse correlations
- The remaining independent variables (EGW share, income, density, employment, etc.) are not target variables, but explanatory variables



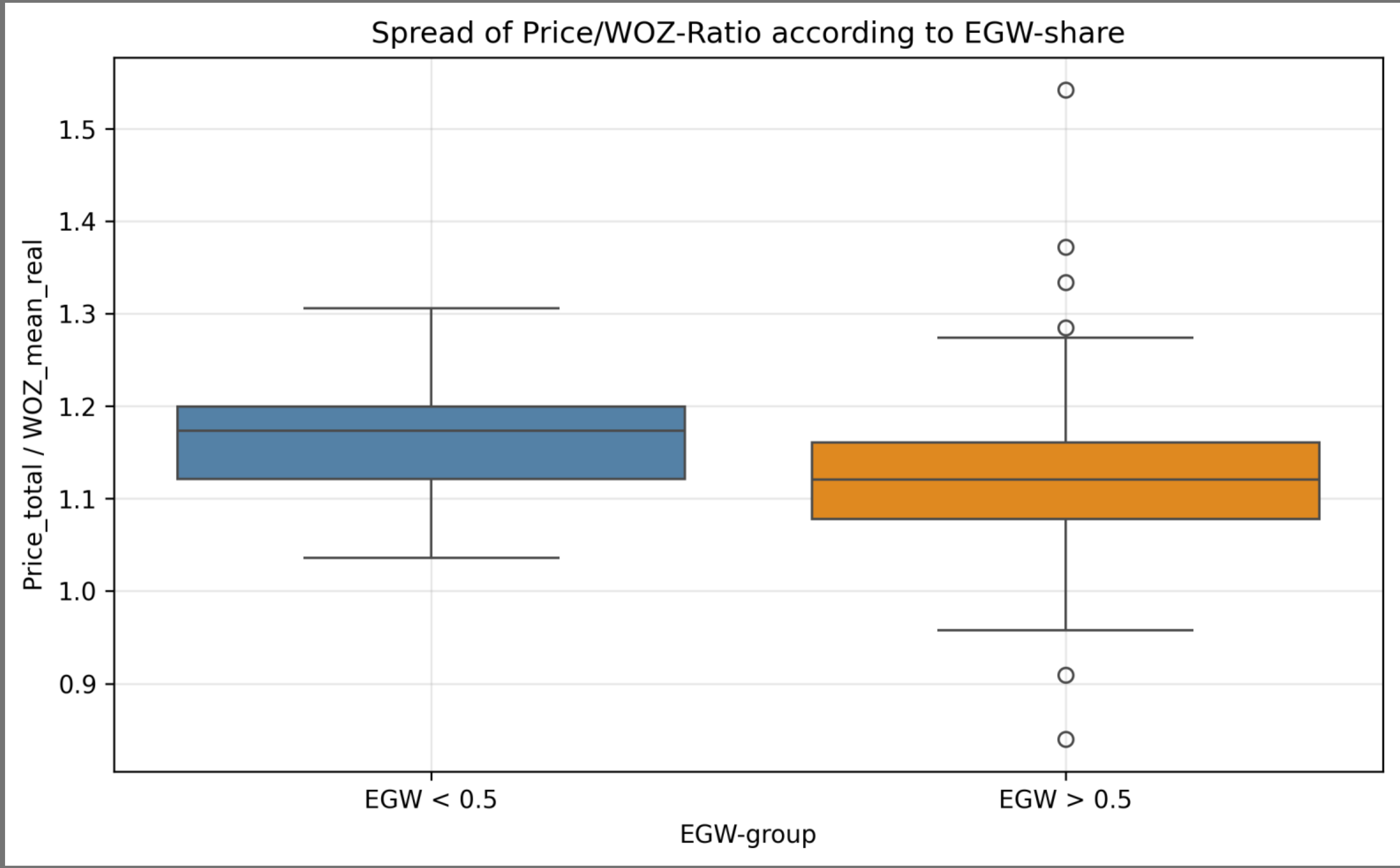
3. EGW and MGW follow different market logics

- **EGW** (single-family houses):
  - more demand driven
  - higher heterogeneity
  - ratio_price_model positive
- **MGW** (multi-family houses):
  - institutional market
  - more influenced by WOZ and regulation
  - ratio_price_model strongly negative
- Models confirm: EGW and MGW should be modeled separately



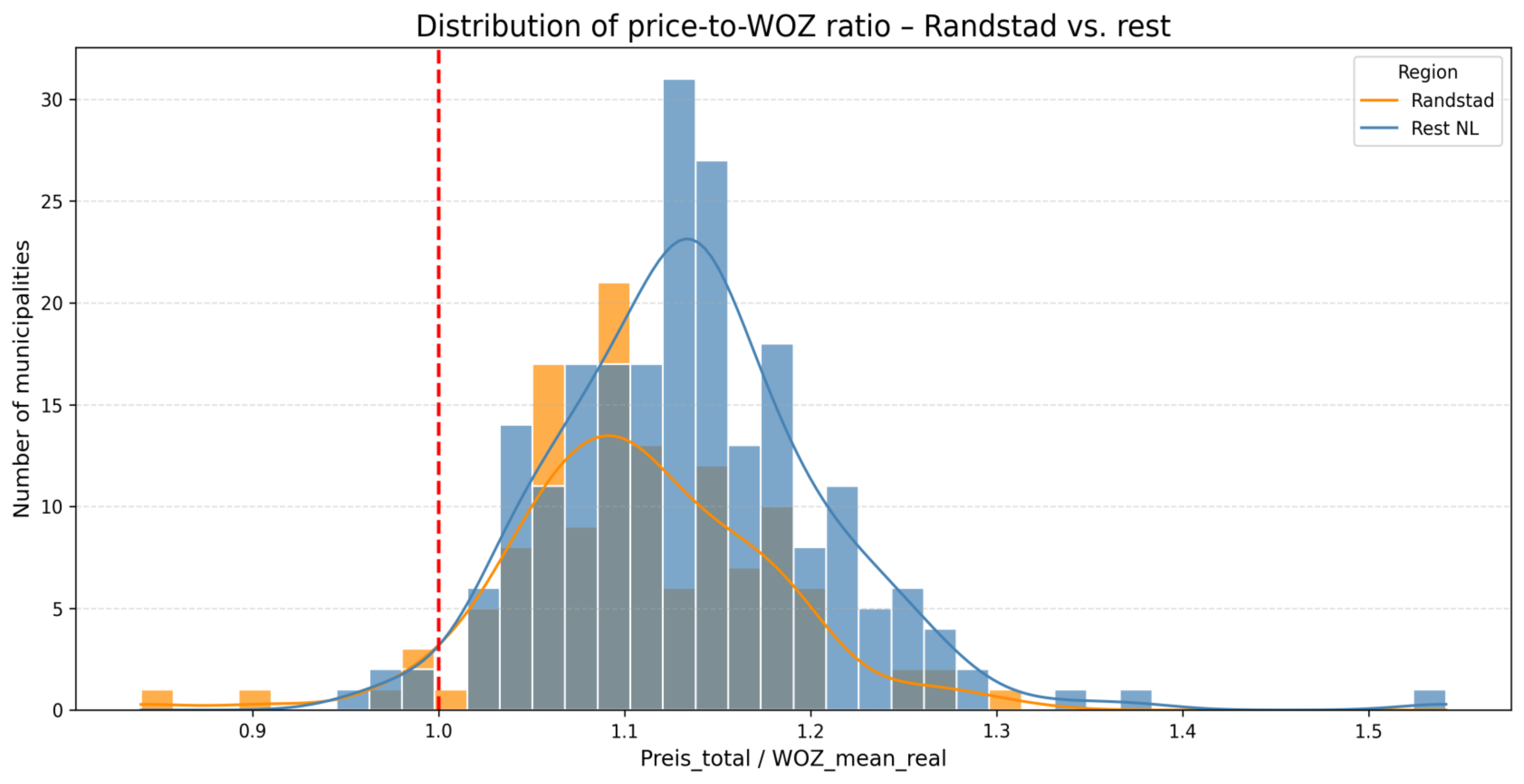
4. EGW does not influence the price level but rather the price dispersion

- OLS and Log-OLS: no effect on the price level
- But: EGW-dominated municipalities have massively higher dispersion
- MGW-dominated municipalities are more stable at the WOZ
- Interpretation:
    - EGW → heterogeneous, micro-location-driven, volatile
    - MGW → standardised, institutional, stable



5. The role of the Randstad as a dynamic sub-market within the Netherlands

- Approximately 45% of the Dutch population lives on around 20% of the country’s land area
- high population density > high proportion of MGW compared to the rest of the NL (30.9% versus 19.6%)
- tendency towards institutional ownership
- The MGW price is based on the annual net rent excluding service charges, as reflected in the WOZ



Conclusion: What does the model explain – and what does it not?

- The model estimates prices well because the WOZ is already a very good estimate
- Structural variables explain only deviations from the WOZ, not the price itself
- EGW and MGW follow different market mechanisms
- The Randstad remains a special case: more dynamic, a higher proportion of MGW properties, and more expensive than the rest of the Netherlands
- The municipal level is too coarse for fine-grained price drivers > breaking it down to postcode or neighbourhood level is the next step
- The model is not a classic hedonic pricing model, but a WOZ-based structural model

QUESTIONS?

Ready for a dialogue about data-driven asset management.

CONTACT VIA LINKEDIN

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This analysis and the underlying portfolio simulation model have been developed for educational, methodological, and professional discussion purposes only.

The calculations are based on simulated, fictitious data structures designed to reflect the core technical mechanisms under Dutch laws and regulations. Although all regulatory formulas and the programming logic have been researched and implemented with the utmost care, this document does not constitute formal legal, tax or financial advice.

Real-world portfolio outcomes depend heavily on specific asset-level variables, including local WOZ-valuations, municipal enforcement policies, and individual tenancy agreements. For asset-specific investment strategies or compliance audits, always consult with a certified Dutch housing law expert or legal counsel.